

Unit7:Tools and Technologies in Data Science

Jupyter Notebook

A Jupyter Notebook is an interactive tool used to write and run code in a web browser.

It allows you to combine:

- ▶ Code
- ▶ Text explanations
- ▶ Outputs (graphs, tables, results)

It works like a digital lab notebook.

Key Features

- ▶ Run code in small blocks (cells)
- ▶ Immediate output display
- ▶ Mix code and documentation
- ▶ Supports visualization (charts, plots)

How it is Used in Data Science

Typical workflow:

- ▶ Load data (CSV, datasets)
- ▶ Clean and preprocess data
- ▶ Analyze patterns
- ▶ Visualize results
- ▶ Build machine learning models

Example Workflow

- ▶ Import libraries (pandas, numpy, matplotlib)
- ▶ Load dataset
- ▶ Perform analysis

Example

```
data = pd.read_csv("data.csv")
```

Why Jupyter Notebook is Useful

- ▶ Easy for beginners
- ▶ Instant results
- ▶ Great for experimentation
- ▶ Widely used in data science and research

Summary

Jupyter Notebook is an interactive environment that combines code, output, and explanation—making it ideal for data science.

What is Version Control?

Version control is a system that helps track changes in files (especially code).

It allows:

- ▶ Tracking changes in files over time
- ▶ Going back to previous versions if something breaks
- ▶ Working collaboratively with others
- ▶ Managing project history safely

Example: Like Google Docs version history, but for programming.

What is Git?

Git is a distributed version control system (DVCS) used to track changes in source code.

Key features:

- ▶ Keeps history of all changes
- ▶ Works offline
- ▶ Fast and lightweight
- ▶ Supports branching and merging
- ▶ Used in almost all software projects

Why Git is important:

- ▶ You can experiment without fear
- ▶ You can undo mistakes easily
- ▶ Multiple developers can work together
- ▶ Keeps project organized

What is GitHub?

GitHub is a cloud-based platform for hosting Git repositories.

Features of GitHub:

- ▶ Remote storage for Git projects
- ▶ Collaboration with teams
- ▶ Code sharing and review
- ▶ Issue tracking
- ▶ Pull requests for teamwork

Git vs GitHub:

- ▶ Git = tool (local)
- ▶ GitHub = service (online)

Difference Between Git and GitHub

- ▶ Git is a software tool
- ▶ GitHub is a cloud platform
- ▶ Git is used for version control
- ▶ GitHub is used to host repositories
- ▶ Git works offline, GitHub needs internet
- ▶ Git was developed by Linus Torvalds
- ▶ GitHub was developed by GitHub Inc.

Basic Git Workflow

Step-by-step workflow:

- ▶ `git init` → initialize repository
- ▶ `git status` → check status of files
- ▶ `git add filename` → add files to staging area
- ▶ `git commit -m "message"` → save snapshot
- ▶ `git remote add origin jrepo-url` → connect GitHub
- ▶ `git push -u origin main` → upload to GitHub

Important Git Commands

- ▶ `git init` → start a repository
- ▶ `git add` → stage changes
- ▶ `git commit` → save changes
- ▶ `git status` → check current changes
- ▶ `git log` → view history of commits
- ▶ `git push` → upload to GitHub
- ▶ `git pull` → download updates

Branching in Git

Branching allows working on different features separately.

- ▶ main branch → stable code
- ▶ feature branch → new development work

Commands:

- ▶ `git branch feature1` → create branch
- ▶ `git checkout feature1` → switch branch

Merging

Merging combines changes from different branches.

- ▶ Used to combine feature branch into main branch
- ▶ Helps in team collaboration
- ▶ Keeps project updated

Command:

- ▶ `git merge feature1`

GitHub Collaboration Tools

- ▶ Pull Requests → propose and review changes
- ▶ Issues → track bugs and tasks
- ▶ Forking → copy another repository

Real-Life Use of Git and GitHub

Used in:

- ▶ Software development
- ▶ Data science projects
- ▶ Open-source contributions
- ▶ Team collaboration
- ▶ Research projects

Summary

- ▶ Git tracks changes in code
- ▶ GitHub stores Git repositories online
- ▶ Git = version control system
- ▶ GitHub = collaboration platform

Together, they are essential tools for modern software development.

What is a Cloud Platform?

A cloud platform is an online environment that allows users to:

- ▶ Write and run code in a browser
- ▶ Store and process data on remote servers
- ▶ Avoid installing heavy software locally

Key Idea: Cloud computers are used instead of your personal computer.

Why Cloud Platforms are Used in Data Science

Cloud platforms are important because they:

- ▶ Provide free CPU/GPU access
- ▶ Support large datasets
- ▶ Enable collaboration
- ▶ Work on low-end devices
- ▶ Require no installation

Google Colab

Google Colab is a cloud-based Jupyter notebook environment.
It is:

- ▶ Free to use
- ▶ Runs Python in browser
- ▶ No installation required
- ▶ Integrated with Google Drive
- ▶ Supports GPU and TPU

Features of Google Colab

- ▶ Pre-installed libraries (NumPy, Pandas, TensorFlow)
- ▶ Easy sharing like Google Docs
- ▶ Supports Jupyter Notebook format
- ▶ Cloud storage support

Used widely in: Machine Learning and AI projects

Advantages and Limitations of Colab

Advantages:

- ▶ Free GPU access
- ▶ Easy to use
- ▶ No setup required

Limitations:

- ▶ Session timeout
- ▶ Internet required
- ▶ Limited memory

Kaggle

Kaggle is a data science platform for learning and competitions.
It provides:

- ▶ Datasets
- ▶ Online notebooks
- ▶ Machine learning competitions
- ▶ Community discussions

Features of Kaggle

- ▶ Free Jupyter-like notebooks
- ▶ Built-in datasets
- ▶ GPU/TPU support
- ▶ Real-world competitions
- ▶ Strong data science community

Advantages and Limitations of Kaggle

Advantages:

- ▶ Free datasets
- ▶ Great for practice
- ▶ Portfolio building

Limitations:

- ▶ Internet required
- ▶ Limited runtime
- ▶ Not for production systems

Google Colab vs Kaggle

- ▶ Colab: Best for learning and experimentation
- ▶ Kaggle: Best for datasets and competitions
- ▶ Both support Python and ML libraries
- ▶ Both run in browser with free compute power

Real-Life Uses and Summary

Used in:

- ▶ Machine Learning projects
- ▶ Data analysis
- ▶ AI research
- ▶ Student assignments

Summary: Cloud platforms like Google Colab and Kaggle allow users to run code, analyze data, and build ML models without installing software locally.

What is Big Data?

Big Data refers to extremely large and complex datasets that cannot be processed using traditional tools.

5 Vs of Big Data:

- ▶ Volume - large amount of data
- ▶ Velocity - fast data generation
- ▶ Variety - different data types
- ▶ Veracity - data quality and reliability
- ▶ Value - useful insights from data

What are Big Data Tools?

Big data tools are frameworks used to:

- ▶ Store large datasets
- ▶ Process distributed data
- ▶ Perform parallel computing
- ▶ Analyze large-scale data

Main tools:

- ▶ Apache Hadoop
- ▶ Apache Spark

Apache Hadoop

Hadoop is an open-source framework for distributed storage and processing of big data.
It is designed to work across clusters of computers.

Hadoop Components

- ▶ HDFS (Hadoop Distributed File System) - stores data in blocks across machines
- ▶ MapReduce - processes data in map and reduce steps
- ▶ YARN - manages resources in cluster
- ▶ Hadoop Common - shared utilities and libraries

Features of Hadoop

- ▶ Highly scalable (handles petabytes of data)
- ▶ Fault tolerant
- ▶ Distributed storage system
- ▶ Cost effective

Limitations of Hadoop

- ▶ Slow due to disk-based processing
- ▶ Not suitable for real-time processing
- ▶ Complex programming (MapReduce)

Apache Spark

Spark is a fast distributed computing system for big data processing and analytics.
It uses in-memory computing for speed.

Spark Components

- ▶ Spark Core - main processing engine
- ▶ Spark SQL - structured data queries
- ▶ Spark Streaming - real-time data processing
- ▶ MLlib - machine learning library
- ▶ GraphX - graph processing

Features of Spark

- ▶ Very fast (in-memory processing)
- ▶ Supports real-time analytics
- ▶ Easy APIs (Python, Java, Scala, R)
- ▶ Works with Hadoop ecosystem

Limitations of Spark

- ▶ Requires high memory (RAM)
- ▶ More expensive infrastructure
- ▶ Complex for beginners

Hadoop vs Spark

- ▶ Hadoop: disk-based processing, slower
- ▶ Spark: in-memory processing, faster
- ▶ Hadoop: batch processing
- ▶ Spark: real-time + batch processing
- ▶ Spark is easier and more modern

Relationship Between Hadoop and Spark

- ▶ Spark can run on Hadoop clusters
- ▶ Hadoop provides storage (HDFS)
- ▶ Spark provides fast processing
- ▶ They are often used together

Real-Life Applications

- ▶ Social media analytics
- ▶ Banking fraud detection
- ▶ Healthcare data analysis
- ▶ E-commerce recommendations
- ▶ IoT data processing

Summary

- ▶ Big Data = large and complex datasets
- ▶ Hadoop = storage + batch processing
- ▶ Spark = fast + real-time processing
- ▶ Both are widely used in industry

Final idea: Spark is faster and more advanced, but Hadoop is still important for storage.

Thank You!